



**Verified Carbon
Standard**

INSTALLATION OF HIGH EFFICIENCY WOOD BURNING COOKSTOVES IN GHANA

Document Prepared by

C-Quest Capital CR Stoves Private Limited

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1 PROJECT DETAILS

1.1 Summary Description of the Project

A summary description of the technologies/ measures to be implemented by the project.

The project involves distribution of fuel-efficient improved cookstoves (ICS) in Ghana. The ICS disseminated through this project will replace the baseline cookstoves i.e., open fire or three stone fires.

Through this project, the distribution and installation of approximately 500,000 ICS will be undertaken for households in Ghana. It is intended that under this project single pot, TLC Rocket Stoves will be distributed, henceforth referred as TLC-CQC Rocket Stoves. The project proponent has implemented several projects in Sub-Saharan Africa and hence has understanding of the cooking cultures prevalent in the region. It is commonly observed that end users tend to return to traditional cookstoves owing to several reasons such as socio-cultural inclination towards open fire stoves as well as its suitability for preparing traditional meals. One of the factors affecting above is also the cooking time. For the rural population in Ghana as with other countries in the region, the women of the household are more concerned about cooking a meal in less time to be able to get back to other household chores and hence use more than one open fire stove for preparing meals. The use of multiple three stone fire stoves in the households for preparing meals to save time is also established by background research and published literature¹ carried out in the host country, Ghana. Each household will receive up to 2 ICS per household to meet their cooking requirements. The ICS will burn wood more efficiently thereby improving thermal transfer to pots, hence saving fuel. Annual quantity of woody biomass used by improved cookstoves will be monitored as mentioned in section 5.2. Improved combustion of fuel reduces the smoke produced and this reduces the health hazards from indoor air pollution. Quantity of firewood saved due to ICS will result in women and children will have to spend less time collecting firewood and reduce the rapidly progressing deforestation in Ghana.

In this grouped project, each ICS constitutes a single project activity instance which is in line with the definition stated in VCS Program Definitions version 04.2.

After dissemination of cookstoves, the end user will be informed in advance that the carbon credits generated through use of ICS will belong to the project promoter generates carbon finance which in turn is used for waving away the cost or highly subsidizing the price of ICS and for recovering project implementation costs.

¹ Rural-urban differences in cooking practices and exposures in Northern Ghana - IOPscience (<https://iopscience.iop.org/article/10.1088/1748-9326/aa7036>)

The location of the project.

The project will be implemented in Ghana. The details of the project location are provided in Section 1.12.

An explanation of how the project is expected to generate GHG emission reductions or removals.

The ICS will substitute the current practice of cooking on open fire stoves. The ICS burns wood more efficiently due to its improved design leading to proper fuel combustion, and more efficient thermal transfer to pots, hence saving fuel and lowering greenhouse gas emissions.

A brief description of the scenario existing prior to the implementation of the project.

The baseline scenario is the continued use of non-renewable wood fuel by the target population to meet similar thermal energy needs as provided by project cookstoves in absence of project activity.

An estimate of annual average and total GHG emission reductions and removals.

The average annual GHG emission reduction from the project is expected to be 1,126,602 t CO₂e and total 11,266,021, tCO₂e over the entire crediting period of 10 years.

Audit Type	Period	Program	VVB Name	Number of years
Validation	24-March-2023 (Date of listing of VCS PD) ²	VCS	Earthood Services Pvt. Ltd.	10

1.2 Sectoral Scope and Project Type

The project is categorised under type/category as below:

- a) **Sectoral scope:** 03 - Energy demand
- b) **Type:** II – Energy efficiency improvement projects

The project is a non-AFOLU grouped project.

1.3 Project Eligibility

The project involves energy efficient cookstove distribution which falls under the category of efficiency improvements in thermal applications, therefore it is eligible under the scope of VCS Program.

1.4 Project Design

The project is a grouped project.

² The project is under registration, as and when the project gets registered PP will update the date.

Eligibility Criteria

For the inclusion of new project activity instances, the project proponent shall ensure that it meets the eligibility criteria below.

S.No.	Criterion	Compliance Requirement	Method of Evaluation
1	Methodology: Meet the applicability conditions set out in the methodology applied to the project	The project activity instance shall use VCS approved methodology-VMR0006: Methodology for Installation of High Efficiency Firewood Cookstoves, Version 1.1 and shall meet the applicability conditions set out in the methodology.	Details of how the new project activity instances meet the applicability conditions can be confirmed in Section 3.2 of this VCS-PD.
2	Technology: Use the technologies and measures specified in the project description	The project activity instance shall implement TLC-CQC Rocket Stoves having minimum efficiency of 25%; below which it shall not be eligible under this grouped project.	Thermal Efficiency test report/ manufacturer 's specification shall be used for establishing minimum efficiency criteria in the first year of stove installation.
3	Baseline Scenario: Are subject to the baseline scenario determined in the project description for the specified project activity and geographic area.	All new project activity instances will be installed within Ghana only and the baseline scenario is the continued use of non-renewable wood fuel by the target population to meet similar thermal energy needs as provided by the project cookstoves in absence of the project activity. This also confirms to the applied methodology.	Baseline scenario can be confirmed in Section 3.4 of this VCS-PD. Baseline scenario and geographical boundary will be confirmed from registration records cum consent deeds of individual ICS which will contain geographical coordinates, address as well as confirmation by end user (stove owner) stating that they were using traditional three stone fire or mud stove or conventional cook stove without a grate or a chimney

			before the installation of project stove.
4	Additionality: Have characteristics with respect to additionality that are consistent with the initial instances for the specified project activity and geographic area.	Each project activity instance shall demonstrate its compliance with- 1.Regulatory Surplus: The project should not be mandated by any law, statute or other regulatory framework, or for UNFCCC non-Annex I countries, any systematically enforced law, statute or other regulatory framework. 2. Positive List: The inclusion of new project activity instances shall have to comply with positive list that is the ICS are installed/distributed at zero cost to end user and the project activity instance has no other source of revenue but GHG credits.	1. Regulatory Surplus: There is no law, statute or government programme or policy in Ghana mandating the project activity nor is there any systematically enforced law, statute or other regulatory framework for such projects. 2. Positive List: Self-Declaration/individual ICS registration form or any other documentary evidence mentioning that end users were not charged for the ICS.
5	Occur within one of the designated geographic areas specified in the project description	All new project activity instances will be installed within Ghana only.	Project Activity Instance Database containing Geographical Coordinates of each household to which cookstoves are distributed.
6	Ownership: Have evidence of project ownership, in respect of each project activity instance, held by the project proponent from the respective start date of each project activity instance	A default Beneficiary Agreement for end users including the provision that emission reductions generated by the project activity are owned by the Project Proponent will be provided for project activity instance.	Copy of registration form cum consent deed from end user (Stove owner) to Project Proponents regarding emission reduction claims.

7	Start Date	The start date of project activity instance shall be same as or later than the grouped project start date	Project Activity Instance Database containing date of installation of each ICS.
8	Where a capacity limit applies to a project activity included in the project, no project activity instance shall exceed such limit	1. The aggregate energy savings by a single project activity shall not exceed the equivalent of 180 GWh_{th}/year as mentioned in the methodology AMS II G: Energy efficiency measures in thermal applications of non-renewable biomass. 2. Each project activity instance that exceeds one percent of the capacity limit (i.e., 1.8 GWh_{th}) shall be identified. 3. Such instances shall be divided into clusters, whereby each cluster is comprised of any system of instances such that each instance is within one kilometer of at least one other instance in the cluster. Instances that are not within one kilometer of any other instance shall not be assigned to clusters.	1. Project Activity Instance Spreadsheet containing estimation of annual energy saving from each project activity instance (i.e., each project ICS). 2. Project Activity Instance ER Spreadsheet containing estimation of annual energy saving from each project activity instance (i.e., each project ICS). 3. The expected annual energy saving from each project activity instance (i.e., each project ICS) comes out to be 0.00937 GWh_{th}/ TLC-CQC Rocket Stove which is well below 1% (i.e., 1.8 GWh_{th}) of the capacity limit. Therefore, the project activity instances under this grouped project shall not be assigned to clusters.
9.	Double Counting	The Improved Cookstove distributed under this project shall be uniquely identifiable based on the distribution records	Project Activity Instance Database containing unique serial number of each ICS. Each ICS distributed will have corresponding end user (stove owner) details (i.e.,

			name, geographical coordinates, address, unique serial number). The combination of this unique serial number of each ICS along with geographical address of the stove owner confirms the unique identity of each project device.
10	Target Population	End user for each project activity instance shall be households, with non-renewable biomass on inefficient traditional mud stove or three stone fire with no improved combustion air supply or flue gas ventilation, that is without a grate or a chimney. Each ICS will be assigned a unique serial number with name of ICS user, address, GPS of household, stove model, distribution date, etc	Target population will be the households relying on non-renewable woody biomass on three-stone fire (not charcoal), or a conventional device with no improved combustion air supply or flue gas ventilation, that is without a grate or a chimney. Database/ICS Registration form if distribution has started or Self- declaration by project participant.
11	Crediting period	Be eligible for crediting from the start date of the instance through to the end of the project crediting period (only). Note that where a new project activity instance starts in a previous verification period, no credit may be sought for GHG emission reductions or removals generated during a previous verification period (as set out in Section 3.4) and new instances are eligible for crediting from the start	The start date of the instance will be the date, the stove began generating GHG emission reductions or removals. The end of the crediting period of instance will be the end date of project crediting period as mentioned in section 1.9.

		of the next verification period.	
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1.5 Project Proponent

Organization name	C-Quest Capital CR Stoves Private Limited
Contact person	Ken Newcombe
Title	Director
Address	38 Beach Road #29-11, South Beach Tower, Singapore (189767).
Telephone	+1-202-247-7976
Email	cqc-operations@cquestcapital.com

1.6 Other Entities Involved in the Project

At the present, C-Quest Capital CR Stoves Private Limited is the sole entity involved in the project.

1.7 Ownership

The project ownership is with C-Quest Capital CR Stoves Private Limited.

During the registration of ICS, the participating household will sign a declaration to transfer the ownership rights of the carbon assets generated from this project to the project proponent. This is in line with clause 5 of section 3.7.1. of VCS project standard version 04.4.

1.8 Project Start Date

27-September-2023 (The date of commissioning of first project activity instance)

1.9 Project Crediting Period

27-September -2023 to 26-September -2033, ten years fixed crediting period.

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

The estimated annual GHG emission reductions/removals of the project are:

☐ <20,000 tCO₂e/year

- ☐ 20,000 – 100,000 tCO₂e/year
- ☐ 100,001 – 1,000,000 tCO₂e/year
- ☒ >1,000,000 tCO₂e/year

Project Scale	
Project	
Large project	X

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
Year 1 27-September -2023 to 26-September -2024	448,625
Year 2 27-September -2024 to 26-September -2025	884,278
Year 3 27-September -2025 to 26-September -2026	1,307,153
Year 4 27-September -2026 to 26-September -2027	1,717,441
Year 5 27-September -2027 to 26-September -2028	1,523,040
Year 6 27-September -2028 to 26-September -2029	1,350,548
Year 7 27-September -2029 to 26-September -2030	1,197,506
Year 8 27-September -2030 to 26-September -2031	1,061,729
Year 9 27-September -2031 to 26-September -2032	941,276
Year 10 27-September -2032 to 26-September -2033	834,425
Total estimated ERs	11,266,021
Total number of crediting years	10

Average annual ERs

1,126,602

1.11 Description of the Project Activity

The project involves distribution of fuel-efficient improved cookstoves (ICS) to replace the baseline cookstoves in households.

The ICS to be deployed under this project is TLC-CQC Rocket Stove which substantially reduces fuel consumption and emissions for conducting cooking and water heating tasks in homes. The ICS improve the efficiency of combustion and thermal transfer to the pot compared with a traditional pot support or three-stone fire by incorporating a number of cutting-edge components, including one or more of; a 'rocket elbow'; a highly insulated combustion chamber which provides a conducive environment for clean and efficient combustion of wood. It substantially reduces wood fuel consumption compared with a three-stone fire or traditional pot support.

Technology/Measure

TLC-CQC Rocket Stove uses a total of 15 bricks that will be made by the households using locally available clay. The average size of the brick would be 22cm x 11cm x 6.5cm. The bricks will be joined together using a mixture of 3 kg clay, 3 kg sand, 3 kg manure/cow dung/ash and 3 liters of water. This ensures reduction in heat loss and better insulation. Metal components have been added to the design to optimize combustion and heat transfer.



TLC-CQC Rocket Stove

Stove components

TLC-CQC stove has a metal top that allows the pot to sit higher improving the flow of air into the combustion chamber and out through the top of the stove.

An adjustable metal pot skirt ensures more effective transfer of heat from the fire into the pot, increasing efficiency and helping to block wind.

TLC-CQC Rocket Stoves come with a metal stick support which is placed in front of and slightly into the opening of the stove and acts as a firewood feeding platform. This ensures adequate airflow while feeding the fuel into the combustion chamber resulting in complete combustion of wood.

According to independent stove efficiency tests performed by Aprovecho Research Centre on the TLC-CQC Rocket Stove, the WBT results yielded an average thermal efficiency of 34.5%. for boiling 5 litres of water.

S.No.	Particular	Technical Specifications / Details
1	Stove Size	Depth: 34.5 cm Width: 34.5 cm Height: 28 cm
2	Efficiency	34.5%
3	Average lifetime	10 years
4	Existing installed capacity	Currently no stoves installed
5	Forecasted installed capacity	500,000 stoves

Note: TLC-CQC Rocket stove specifications are proposed for the present project activity. However, the PP at their sole discretion or at end user(s) request may modify / update stove model with similar specifications. If the stove models are different from the present specifications, then the technical details of variant, including efficiency value, will be included in the monitoring report during the respective monitoring period. The changes in the stove model will not impact the applicability of the methodology, additionality, or the appropriateness of the baseline scenario as mentioned under section 1.4 (eligibility criteria) of this PD.

The life of TLC stove is at least 10 years owing to it being a clay and brick stove which can be easily repaired using a mixture of local clay, cow dung/ash, sand and water etc. on a regular basis. The metal parts used in the stoves are under extended warranty from PP/manufacture side throughout the projected life span and will be replaced in case of request received from the stove owner for replacement. PP has confirmed the life span of 10 years through the above measures.

Baseline cookstove

It is a common and traditional practice in the project area to use three stone fired stoves or a conventional device with no improved combustion air supply or flue gas ventilation, that is without a grate or a chimney. There is no other type of stove or technology being used or observed in the project area³. Hence, the efficiency for three stones fired or conventional stoves stands at 0.1 or 10% as per the applied methodology and the same is considered in the baseline estimations.

Data collection of ICS end-user

Project proponent must gather the necessary information to identify households using its ICS during the course of the project. To facilitate this process, each ICS will be assigned a unique serial number. This number will be recorded during the registration process together with the following information (as appropriate and as available):

- Name of ICS user or head of the household

³ Ghana Background research Report v1

- Address / GPS of ICS household
- Phone number of ICS user or household, where available.
- Stove model
- Date of distribution/installation
- ICS serial number
- Retailer/distributor information

The information collected will be stored in the electronic database which will serve as project database for project monitoring and sampling purposes.

Unique identification of End users

A unique ID comprising of a combination of end-user name, geographical location as well as a serial number will be used to identify the ICS end users of this project activity. This identification mechanism will help the PP to differentiate VCUs specific to each project activity that will be undertaken in the project boundary. The PP will ensure that double accounting is not done for the projects and the ICS registered to the current project activity are used to claim VCUs.

Ex-ante v/s Actual emission reductions

The ex-ante emission reductions have been calculated as demonstrated in section 4.4. However, the actual emissions reductions during the verification will depend on the values of $B_{y=1,new,i,j,survey}$ and $N_{y,i,j}$ obtained during the monitoring surveys. Also, the projected stove installation might get extended over a year or maybe even two years depending on the rate of construction of the stoves on field and therefore for first few years the emission reduction could be lower than the ex-ante calculated values. The uncertainty in number of stoves installed and operational during the monitoring period will primarily result in the differences of the ex-ante and actual emission reductions achieved during the crediting period.

1.12 Project Location

The project location will be the geographical boundary of Republic of Ghana with coordinates 7°56'47.5" N latitude and 1°1.392' W longitude.⁴

The geodetic coordinates for the first TLCRS stove (first project activity instance) installed under this grouped project is 5.4079185 latitude and -0.9170353 longitude.

⁴ <https://www.geodatos.net/en/coordinates/ghana>


Ghana map⁵


Map of project area (KML file shared with VVB)

1.13 Conditions Prior to Project Initiation

The conditions prior to project initiation is the continued use of non-renewable wood fuel (firewood) by the target population to meet similar thermal energy needs as provided by project cookstoves in absence of project activity. The pre-project devices are inefficient three-stone fires or conventional systems lacking improved combustion air supply mechanism and flue gas ventilation system, i.e., traditional stoves.

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

There are no laws and regulations governing the use of improved cookstoves in Ghana households. The project is a voluntary effort by the project proponent.

1.15 Participation under Other GHG Programs

1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

The project has not been registered, nor is it seeking registration under any other GHG program.

⁵ <https://www.ghanamissionun.org/map-regions-in-ghana/>

1.15.2 Projects Rejected by Other GHG Programs

The project has not been rejected by any other GHG program.

1.16 Other Forms of Credit

1.16.1 Emissions Trading Programs and Other Binding Limits

The project is not included in an emissions trading program or any other mechanism that includes GHG allowance trading.

1.16.2 Other Forms of Environmental Credit

The project has not sought or received another form of GHG-related environmental credit.

Supply Chain (Scope 3) Emissions

As per Section 3.23.9 of the VCS Standard, v4.4, the “producer(s) or retailer(s) of the impacted good or service are known but not involved in the project or do not have a website”. The Project Proponent has informed the manufacturer of the TLC-CQC Rocket Stoves as well as the implementing partner in Ghana at the time of signing agreement for the goods and services provided by them, that ICS will be distributed in Ghana under this project activity. Also, intimated them about the project registration under VERRA for the generation of VCUs.

A public statement(s) by the Project Proponent has been made throughout the project crediting period on its website for reference. The impacted good or service's producer(s) have been notified of the project and the potential risk of Scope 3 emissions double claiming via email. Evidence of the public statement(s) is provided in this report as appendix C during validation process.

1.17 Sustainable Development Contributions

The project contributes to sustainable development in a number of ways:

a. Environmental Sustainability

- The project will help significantly reduce greenhouse gas emissions over its lifetime.
- The project will help reduce the use of non-renewable biomass from forests, thus assist in conserving existing forest stock and the protection of natural forest eco-systems and wildlife habitats.

b. Social Sustainability

- Considerably less time need to be spent collecting wood fuel for the family home thereby reducing the work burden on rural families and presenting alternative opportunities for economic development.
- The amount of indoor pollutants from the burning of biomass in the family home will be reduced. Less carbon dioxide, carbon monoxide and particulates will be emitted due to

the decrease in total biomass burned and an increase in the temperature of combustion.

- The stove provides a safer method for combusting biomass for cooking, helping to reduce burn injuries, especially for children, in the family home.

c. Economic Sustainability

- The project will help develop a section of the local economy, in the distribution, local assembly, maintenance and monitoring activities.
- Household expenditures on cooking fuel will be reduced through the use of the ICS.
- Saved household labour can be diverted to more productive economic activities.
- The project will create local employment opportunities in operational and management roles, as well as future assembly and/or manufacturing initiatives.

Following SDGs are targeted through this project

SDG 3: Good Health and Well-being

Reduce fine particulate matter during cooking to improve health and reduce burdens of disease.

SDG 4: Quality Education

Contribute to community & individual awareness of climate change & sustainable development.

SDG 5: Gender Equality

Reduce drudgery for women and children, by reducing time spent collecting firewood and cooking.

SDG 7: Affordable and Clean Energy

Access to clean cooking technologies.

SDG 8: Decent Work and Economic Growth

Employ country national staff, with focus on women.

SDG 13: Climate Action

Emission reduction per stove over 10-year project lifecycle.

SDG 15: Life on Land

Reduce deforestation from land surrounding project areas.

1.18 Additional Information Relevant to the Project

Leakage Management

Not applicable as the project adopts a net gross adjustment factor of 95% to account for leakage.

Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

Further Information

Not applicable. There are no laws and regulations governing the use of improved cookstoves in Ghana households. The project is a voluntary effort by the project proponent.

2 SAFEGUARDS

2.1 No Net Harm

The project will only bring positive impacts on environmental and socio-economic aspects as elaborated in Section 1.17. No potential negative environmental or socio-economic impacts have been identified for the project.

2.2 Local Stakeholder Consultation

The National Stakeholder Consultation with local stakeholders on C-Quest Capital clean cookstoves in Ghana, was convened by C-Quest Capital as part its engagement process. The stakeholder's consultation was held on Monday 03-April-2023 at the University of Ghana, Accra-Ghana. Newspaper announcement was published in Daily Graphic on 17-March-2023 [see Appendix A]. E-mail invitations were sent from C-Quest Capital on 22-March-2023 to people from organizations around the country including government, NGOs, stove manufacturers, project developers, academia and private and public-sector entities involved with social work, cookstoves and/or sustainable energy. The email included an Agenda, Non-technical summary, and the feedback form. Invitation letter was sent to the CEO of The Ghana Alliance for Clean Cook stoves and Fuels (GHACCO) on 22-March-2023.

The meeting was attended by 107 members in person and 08 members remotely through Microsoft teams including representatives of government, NGOs, stove manufacturers, project developers, academia, private and public-sector entities involved with social work, cookstoves and/or sustainable energy. Mr. Alfred Adjabeng, CQC Ghana Country Manager, provided presentation with vital education to participants about C-Quest Capital and overview of Verified Carbon Standards & SD-VISta and Ghana C-Quest Projects. The participant has overall positive feedback regarding the project activity. Few members have raised query that why the cookstoves will be issued free. Mr. Alfred has responded that, project is using carbon finance for the successful operation of this project hence the PP is providing free stoves. The question was asked whether the stoves will be imported or sourced within the country. It has been made clear that the stoves will be made from locally available material like clay, ash, dung, water etc. and metal components will be imported. In future, if feasible the metal parts may be manufactured in Ghana. Overall, the feedback has been received that the project activity should be implemented in rural areas and less privileged members should receive the project stoves. CQC representatives has made it clear that the project will be implemented in rural areas and will be implemented in other region of Ghana in phased manner. Photographs, recording and minutes of meeting are provided to VVB during validation process.

Ongoing communication

The local team / ground staff of the PP will be in regular contact with the end users, Village chief / local leadership throughout the life of the project activity. The local team can provide necessary

Grievance Redressal mechanism

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graph TD; BC[Beneficiary] --> GSC[Ground Staff/Field Coordinator]; GSC --> IP[Implementation Partner]; IP --> OM[Operational Manager]; OM --> COH[Country Operational Head/ Project Head/ Lead]; COH --> CQC[CQC Head Office]; BC -.-> VC[Village Chief]; VC -.-> GSC; VC -.-> BC;
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The flowchart illustrates the traditional approach for community-based management of health services. It shows a vertical hierarchy of roles: Beneficiary, Ground Staff/Field Coordinator, Implementation Partner, Operational Manager, Country Operational Head/ Project Head/ Lead, and CQC Head Office. A Village Chief is shown as an external entity that interacts with the Beneficiary and the Ground Staff/Field Coordinator. The flow of the traditional approach is indicated by a dashed line connecting the Village Chief to the Ground Staff/Field Coordinator.

The Grievance Policy and Procedure can be accessed through a phone number of the PP/implementing partner office in Ghana, contact number is also available on the brochure provided to each household, which is the most accessible manner to provide feedback. PP policy and procedure outline the process to evaluate and respond to complaints; the procedure details when a serious level complaint is received it should be communicated to PP country teams and then it would be best addressed by the implementing partner.

No negative environmental impacts have been identified from the project and environmental impact assessment (EIA) is not required for the project.

The project was open for public comments from 24-March-2023 to 23-April-2023. However, no comments were received during this period.

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This section is not applicable as the project is a non-AFOLU project.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

1. VMR0006: Methodology for Installation of High Efficiency Firewood Cookstoves, Version 1.1⁶
2. Tool 30: "Calculation of the fraction of non-renewable biomass" (version 4.0⁷)
3. Standard: "Sampling and Surveys for CDM project activities and programmes of activities" (version 09.0⁸)

3.2 Applicability of Methodology

Applicability criterion	How the project complies
Project activities must introduce efficiency improvements in thermal applications of non-renewable biomass in the following premises: households, community-based kitchen, institutions (e.g., schools) or small and medium-sized enterprises (SMEs).	The proposed project involves deployment of Improved cookstoves which leads to improvement in thermal application of firewood (non-renewable biomass) in household premises.
The project stove shall have specified high-power thermal efficiency of at least 25% per the manufacturer's specifications and shall exclusively use woody biomass and can be single pot or multi-pot;	TLC-CQC Rocket stoves planned to be installed under this project are single pot wood cookstoves that have an efficiency of 34.5% as per the manufacturer's specifications.
Non-renewable biomass has been used in the project region since 31-December-1989, using survey methods or referring to published literature, official reports or statistics;	Non-renewable biomass has been used since 31-December-1989 in Ghana as demonstrated at below.
For the specific case of biomass residues processed as a fuel (e.g. briquettes, wood chips). it shall be demonstrated that: (a) It is	Not applicable. The ICS is introduced as energy efficiency measure to replace baseline

⁶ <https://verra.org/wp-content/uploads/2021/07/VMR0006-Methodology-for-Installation-of-High-Efficiency-Firewood-Cookstoves-v1.1.pdf>

⁷ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-30-v4.0.pdf>

⁸ https://cdm.unfccc.int/sunsetcms/storage/contents/stored-file-20210531160756223/Meth_Stan05.pdf

produced using exclusively renewable biomass (more than one type of biomass may be used). (b) The consumption of the fuel should be monitored during the crediting period and (c) Energy use for renewable biomass processing (e.g., shredding and compacting in the case of briquetting) may be considered as equivalent to the upstream emissions associated with the processing of the displaced fossil fuel and hence disregarded.	stoves and reduce the use of non-renewable biomass for combustion.
Both 'Projects' and 'Large Projects' can use this methodology.	The proposed project is expected to generate 1,126,602 tonnes CO ₂ per year which is greater than 300,000 tonnes CO ₂ per year and thus qualifies as a "large project" as per section 3.10.1 of the VCS Standard v4.4.
The PD shall explain the proposed method for distribution of project devices including the method to avoid double counting of emission reductions such as unique identifications of product and end-user locations (e.g., programme logo).	Each ICS in this project will be identified by a unique combination of end username and geographical location, as well as a serial number. The serial number will be a unique number which will allow for a clear distinction between the stoves. No individual serial number can be repeated within the project, thus ensuring that each stove is counted only once in the proposed project.
The PD shall also explain how the proposed procedures prevent double counting of emission reductions, for example to avoid that project stove manufacturers, wholesale providers or others claim credit for emission reductions from the project devices.	The stove manufacturers/wholesale providers/end users shall sign an undertaking stating clearly that the PP or an entity authorized by it shall be the sole owner of the VCU's arising from the project.

Evidence that the non-renewable biomass has been in use since 1989.

Non-renewable biomass has been in use since 31-December-1989 as evidenced by various FAO statistical data. The Global Forest Resources Assessment 2020⁹ (FAO) indicates that between years 1990 and 2020, Ghana lost 1,938,550 hectares of forest areas. Forest area decreased from 9,924,260 ha in year 1990 to 7,985,710 ha in year 2020.

⁹ FAO, Global Forest Resources Assessment 2020, Country Report, Ghana, Page 10
<https://www.fao.org/3/cb0004en/cb0004en.pdf>

Decline in natural forest area between 1990–2020

FRA 2020 report, Ghana

FRA categories	Area (1000 ha)								
	1990	2000	2010	2015	2016	2017	2018	2019	2020
Forest (a)	9 924.26	8 848.61	7 942.93	7 880.44	7 922.24	7 964.04	7 971.26	7 978.48	7 985.71
Other wooded land (a)	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Other land (c-a-b)	12 829.74	13 905.39	14 811.07	14 873.56	14 831.76	14 789.96	14 782.74	14 775.52	14 768.29
Total land area (c)	22 754.00	22 754.00	22 754.00	22 754.00	22 754.00	22 754.00	22 754.00	22 754.00	22 754.00

Thus, it can be concluded that losses have been reported in forest areas from before 1989 thereby hinting that fuelwood used across Ghana comes from non-renewable sources.

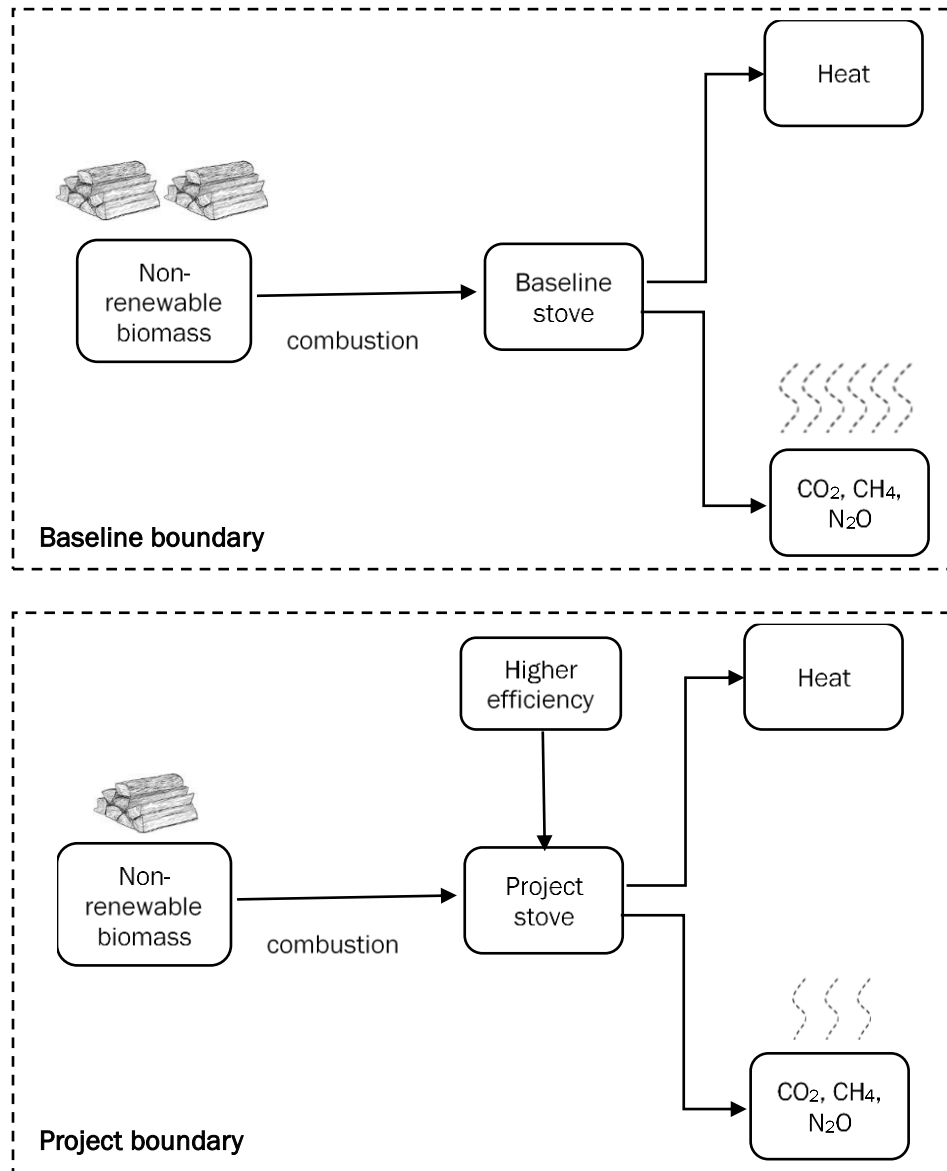
3.3 Project Boundary

As per VMR0006: Methodology for Installation of High Efficiency Firewood Cookstoves, Version 1.1, the project boundary is determined following the procedure provided in CDM methodology AMS-II.G. ver13.0

Component	Project Boundary
The project boundary is the physical, geographical site of the efficient devices that utilize biomass.	The project boundary for this project is the geographical boundary of republic of Ghana. Refer section 1.12 for more details.

Source		Gas	Included?	Justification/Explanation
Baseline	Emission from use of non-renewable biomass/Fossil fuel	CO ₂	Yes	Major source
		CH ₄	Yes	Major source
		N ₂ O	Yes	Major source
		Other	No	No other source identified
Project	Emission from use of non-renewable biomass	CO ₂	Yes	Major source
		CH ₄	Yes	Major source
		N ₂ O	Yes	Major source
		Other	No	No other source identified

A representation of the baseline boundary and project boundary are given as below.



3.4 Baseline Scenario

The baseline scenario is the continued use of non-renewable wood fuel (firewood) by the target population to meet similar thermal energy needs as provided by project cookstoves in absence of project activity. The baseline cookstoves are three-stone fires or conventional systems lacking improved combustion air supply mechanism and flue gas ventilation system i.e., traditional stoves.

It is a common and traditional practice in the project area to use three stone fired stoves or a conventional device with no improved combustion air supply or flue gas ventilation, that is without

a grate or a chimney. There is no other type of stove or technology being used or observed in the project area. This has been confirmed through baseline assessment report conducted¹⁰. Hence, the efficiency for three stones fired or conventional stoves stands at 0.1 or 10% as per the applied methodology and the same is considered in the baseline estimations.

This project activity will replace baseline stoves which are three stone fire stoves¹¹ using large quantities of non-renewable biomass with high efficiency stoves using woody biomass more economically thereby reducing emissions of greenhouse gases. This project intends to target only those households that are using the three stone fire or a conventional device with no improved combustion air supply or flue gas ventilation, that is without a grate or a chimney. At the time of project stove registration this will be confirmed with HH and then only registration will be completed.

3.5 Additionality

The methodology uses activity method for the demonstration of additionality.

Activity Method

Step 1: Regulatory Surplus

There is no mandated government programme or policy in host country of this project ensuring the distribution of domestic fuel-efficient cookstoves. The project is not mandated by any law, statute or other regulatory framework, or for UNFCCC non-Annex I countries, any systematically enforced law, statute or other regulatory framework.

Households may only participate voluntarily in this project. It is hereby confirmed that the proposed project is a voluntary coordinated action by CQC.

Step 2: Positive List

As per Section 3.2, the project meets the applicability conditions of the methodology which represent the positive list.

The project installs the ICS at zero cost to the household and has no other source of revenue other than the sale of GHG credits.

The project is not implemented as part of government schemes or supported by multilateral funds.

Conclusion: As the project fulfills the conditions above, it is deemed additional.

¹⁰ Ghana Background research Report v1

¹¹ During project stove registration the end user confirms to using three stone fire stoves for cooking meals post which they are provided with project stoves.

Step 3: Project Method

For any project activity where stoves are not provided at zero cost to the end-user or has any other source of revenues other than the sale of GHG credits, then the project activity shall apply investment analysis method set out in the CDM Tool for the Demonstration and Assessment of Additionality included in AMS-II.G to determine that the proposed project activity is either: 1) not the most economically or financially attractive, or 2) not economically or financially feasible.

This step is not applicable under project activity because the stoves are provided at zero cost to the end-users.

3.6 Methodology Deviations

The project did not apply any methodology deviations.

4 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

The methodology (VMR0006: Methodology for Installation of High Efficiency Firewood Cookstoves v.1.1) does not account for baseline emissions separately, but instead quantifies net emission reductions achieved by the project. Please refer to Section 4.4.

4.2 Project Emissions

The methodology (VMR0006: Methodology for Installation of High Efficiency Firewood Cookstoves v.1.1) does not account for project emissions separately, but instead quantifies net emission reductions achieved by the project. Please refer to Section 4.4.

4.3 Leakage

Leakage shall be considered as default 0.95 in accordance with methodology (VMR0006: Methodology for Installation of High Efficiency Firewood Cookstoves v.1.1).

4.4 Net GHG Emission Reductions and Removals

The improved cookstove is introduced as energy efficiency measure in the project, therefore equations 1 and 2 of the methodology 'VMR0006: Methodology for Installation of High Efficiency Firewood Cookstoves, Version 1.1' will be applied to calculate the net GHG emission reductions.

$$ER_y = \sum_i \sum_j ER_{y,i,j} \quad \text{Equation (1)}$$

Where:

- i = Indices for the situation where more than one type/model of improved cookstove is introduced to replace traditional stove or three-stone fire
- j = Indices for the situation where there is more than one batch of improved cookstove of type i
- ER_y = Emission reductions during year y in t CO₂e
- $ER_{y,i,j}$ = Emission reductions by improved cookstove of type i and batch j during year y in t CO₂e

$$ER_{y,i,j} = B_{y,savings,i,j} \times NCV_{wood\ fuel} \times f_{NRB,y} \times (EF_{wf,CO_2} + EF_{wf,non\ CO_2}) \times N_{y,i,j} \times 0.95 \quad \text{Equation (2)}$$

Where:

- $B_{y,savings,i,j}$ = Quantity of woody biomass that is saved in tonnes per improved cookstove of type i and batch j during year y
- $f_{NRB,y}$ = Fraction of woody biomass that can be established as non-renewable biomass (fNRB)¹²
- $NCV_{wood\ fuel}$ = Net calorific value of the non-renewable woody biomass that is substituted or reduced (IPCC default for wood fuel, 0.0156 TJ/tonne)¹³
- EF_{wf,CO_2} = CO₂ emission factor for the use of wood fuel in baseline scenario (IPCC default for wood fuel, 112 tCO₂/TJ)¹⁴
- $EF_{wf,non\ CO_2}$ = Non-CO₂ emission factor for the use of wood fuel in baseline scenario (IPCC default for wood fuel, 26.23 tCO₂/TJ)¹⁵
- $N_{y,i,j}$ = Number of improved cookstoves of type i and batch j operating during year y

¹² Third party report by C4Ecosolutions (Pty) Ltd dated 14-December 2022.

¹³ [2006 IPCC Guidelines for National Greenhouse Gas Inventories: Volume 2 Energy, Chapter 1 Introduction](#)

¹⁴ [2006 IPCC Guidelines for National Greenhouse Gas Inventories: Volume 2 Energy, Chapter 2 Stationary Combustion](#)

¹⁵ [2006 IPCC Guidelines for National Greenhouse Gas Inventories: Volume 2 Energy, Chapter 2 Stationary Combustion](#)

0.95 = Discount factor to account for leakage

The quantify of woody biomass saved due to implementation of improved cookstoves to be estimated using equation 4 of the methodology below:

$$B_{y,savings,i,j} = B_{y=1,new,i,survey} \times \left(\frac{\eta_{new,y,i,j}}{\eta_{old}} - 1 \right) \quad \text{Equation (4)}$$

where

- η_{old} = Efficiency of baseline cookstove
- $\eta_{new,y,i,j}$ = Efficiency of the improved cookstove type i and batch j determined through water boiling test (WBT) during year y
Alternatively, efficiency may be determined using Equation 5.
- $B_{y=1,new,i,j,survey}$ = Annual quantity of woody biomass used by improved cookstoves in tonnes per device of type i and batch j , determined in the first year of the implementation of the project through a sample survey.

$$\eta_{new,y,i,j} = \eta_p \times (DF_n)^{y-1} \times 0.94 \quad \text{Equation (5)}$$

where

- η_p = Efficiency of project stove (fraction) at the start of project activity.
- $(DF_n)^{y-1}$ = Discount factor to account for efficiency loss of project cookstove per year of operation (fraction). This value may be based on actual monitoring or based on manufacturer's declaration on expected loss in efficiency or through publicly available literature on relevant industry standards. Alternatively default value of 0.99 efficiency loss per year can be considered.
- 0.94 = Adjustment factor to account for uncertainty related to project cookstove efficiency test.

For ex-ante calculation purpose, the assumption below is applied.

- 1) The project will install up to 500,000 ICS.
- 2) The life span of ICS is 10 years. For conservative estimate of ER, the annual stove loss rate is assumed at 10%.
- 3) $B_{y=1,new,i,j,survey}$ is assumed as 0.96 tonnes / device / year ((This is only an assumed value used for ex-ante calculation based on the literature review (details provided in ER sheet), to be confirmed during first Monitoring Period)
- 4) Fraction of non-renewable biomass (fNRB) calculation for Ghana

Summary of the equations applied to estimate f_{NRB} according to the latest version of the CDM Tool 30 (EB 115 Annex 22 version 4.0 2022).

The fraction of harvested biomass that can be established as non-renewable biomass is the f_{NRB} . This fraction ranges between 0 and 1, where 0 indicates that 100% of harvested biomass is renewable (0% is non-renewable) and 1 indicates that 100% is non-renewable. The following equation is used to calculate f_{NRB} :

$$f_{NRB} = \frac{NRB}{NRB + RB}$$

Where:

f_{NRB} = Fraction of non-renewable biomass

NRB = Non-renewable biomass (t/yr)

RB = Renewable biomass (t/yr)

The quantity of non-renewable biomass (NRB) is the difference between the annual biomass consumption (H) and the quantity of renewable biomass (RB), calculated by the following equation:

$$NRB = H - RB$$

Where:

H = Total woody biomass consumption in the absence of project activity (t/yr)

Renewable biomass is the product of the mean annual increment of woody biomass growth and the total extent of the forest and other wooded areas of the country where wood extraction is not prohibited or geographically remote, calculated by the following equation:

$$RB = \sum(MAI_{forest,i} \times (F_{forest,i} - P_{forest,i})) + \sum(MAI_{other,i} \times (F_{other,i} - P_{other,i}))$$

Where:

MAI = Mean annual increment of woody biomass (t/ha/year)

F_{forest} = Extent of forest in the relevant period (ha)

F_{other} = Extent of other land in the relevant period (ha)

P_{forest} = Extent of non-accessible area within forest areas (ha)

P_{other} = Extent of non-accessible area within other land areas (ha)

i = Sub-category i of forest areas and other land areas

The total woody biomass consumption in a country comprises its domestic and non-domestic energy and non-domestic non-energy consumption. For Ghana, this total is estimated using consumption data sourced from the FAO Forest Products Databases and World Bank population statistics. The reported volumes of fuelwood consumed in cubic meters are converted to metric tonnes using the FAO default conversion factor. Finally, consumed volumes of commercial forest products — including industrial roundwood, sawn wood, veneer sheets and wood-based panels are converted to metric tonnes using a biomass conversion and expansion factor provided for the region.

Table: Total 2021 population, woody biomass consumption, and per capita consumption estimates for Ghana:

Variable	Value	Source
Population		

Urban	17,820,023	The World Bank ¹⁶
Rural	13,252,922	The World Bank
Total	31,072,945	The World Bank
Domestic woody biomass consumption		
Wood fuel consumption (t/yr)	35,278,668	Calculated
Domestic per capita consumption rates		
Wood fuel ((m3/capita/yr)	1.566	FAO10 ¹⁷
Non-domestic (energy) wood consumption		
Wood charcoal (t/yr)	8,451,841	Calculated
Non-domestic (non-energy) wood consumption		
Industrial roundwood (m3/yr)	1,833,304	Calculated
Sawnwood (m3/yr)	403,948	Calculated
Veneer sheets (m3/yr)	248,584	Calculated
Wood-based panels (m3/yr)	217,511	Calculated
Non-domestic per capita consumption rates		
Wood charcoal (t/capita/yr)	0.068	FAO
Industrial roundwood (m3/capita/yr)	0.059	FAO
Sawnwood (m3/capita/yr)	0.013	FAO
Veneer Sheets (m3/capita/yr)	0.008	FAO
Wood-based panels (m3/capita/yr)	0.007	FAO
Conversion factors		
Wood density (t/m3)	0.725	FAO
Biomass conversion and expansion factor (t/m3)	1.300	FAO
Charcoal to wood biomass factor	4	CDM ¹⁸
Total woody biomass consumption (t/yr)	47,244,859	Calculated

Variable	Value	Source
Woody biomass consumption (t/yr)	47,244,859	UN Data, FAO, IPCC
Renewable biomass (t/yr)	4,725,942	Calculated
Non-renewable biomass (t/yr)	42,518,917	Calculated (Equation 2 of CDM tool 30)
Fraction of non-renewable biomass	0.90	Calculated (Equation 1 of CDM Tool 30)

The difference between woody biomass consumption and renewable biomass is considered to be non-renewable (CDM Tool 30, Equation 2). Non-renewable biomass utilization in Ghana is, therefore, estimated to be 42,518,917 t/yr. The fraction of non-renewable biomass is the quotient of the non-renewable and the total biomass (CDM Tool 30, Equation 1). The fraction of non-renewable biomass for Ghana is, therefore, estimated to be 0.90.

- 5) Annual stove loss rate is assumed as 10%.

¹⁶ The World Bank. World Bank Open Data. Accessed on 05/05/2022 from <https://data.worldbank.org/> (2022).

¹⁷ FAO Statistics. Forest Products 2019. (2021)

¹⁸ CDM. Methodological tool 30: Calculation of the fraction of non-renewable biomass. Version 04.0. <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-30-v4.0.pdf> (2022).

Equations used for determination of number of ICS operating during year y

Annual Stove Installation: It is assumed that the PP will install 125,000 stoves every year for first 4 years (i.e. Year1, Year2, Year 3 and Year 4). Total 500,000 stoves will be installed in this project activity. Annual stove loss rate is assumed as 10%. Project Proponent would like to state that the number of project devices operating during year y is a monitored value which can be either measured directly or based on a representative sample. This is as per VMR0006 - Methodology for Installation of High Efficiency Firewood Cookstoves v1.1.

However, the project proponent has followed a conservative approach to calculate estimated net GHG emission reductions. Further Project Proponent would like to state that monitored value of $N(y,i,j)$ based on surveys will be provided at the time of verification under grouped project activity.

The number of ICS operating during year y have been estimated using the below equation:

$$\begin{aligned} N_{y,i,j} &= 125,000 \times [1 - 10\%] \\ &= 125,000 \times 90\% \\ &= 112,500 \end{aligned}$$

Example of calculation:

If $y = 2$,

$$\begin{aligned} N_{y,i,j} &= 250,000 \times [1 - 10\%] \\ &= 250,000 \times 90\% \\ &= 225,000 \end{aligned}$$

Hence, the number of ICS operating during year y is as below:

Year (y)	$N_{y,i,j}$
1	112,500
2	225,000
3	337,000
4	450,000
5	405,000
6	364,000
7	328,050
8	295,245

9	265,721
10	239,148

Determination of efficiency of ICS during year y

$$\eta_{new,y,i,j} = \eta_p \times (DF_n)^{y-1} \times 0.94$$

Where:

$$\eta_p = 34.5\%$$

$$DF_n = 0.99$$

Example of calculation:

If y= 2

$$\eta_{new,y,i,j} = 34.5\% \times (0.99)^{2-1} \times 0.94 = 32.11\%$$

Hence the efficiency of ICS during year y is as below:

Year (y)	$\eta_{new,y,i,j}$
1	32.43%
2	32.11%
3	31.78%
4	31.47%
5	31.15%
6	30.84%
7	30.53%
8	30.23%
9	29.92%
10	29.63%

Determination of quantity of woody biomass that is saved in tonnes per ICS during year y

The Project Proponent would like to state that actual value of $B_{y=1,new,i,j,survey}$ will be determined in the first year of the implementation of the project through a sample survey.

$$B_{y,savings,i,j} = B_{y=1,new,i,survey} \times \left(\frac{\eta_{new,y,i,j}}{\eta_{old}} - 1 \right)$$

Example of calculation:

If $y=2$,

$$B_{y,savings,i,j} = 0.96 \times [(0.3211/0.1) - 1] = 2.132 \text{ tonnes}$$

Year (y)	$B_{y=1,new,i,survey}$	$\eta_{new,y,i,j}$	η_{old}	$B_{y,savings,i,j}$
1	0.96	32.43%	0.1	2.163
2	0.96	32.11%	0.1	2.132
3	0.96	31.78%	0.1	2.101
4	0.96	31.47%	0.1	2.070
5	0.96	31.15%	0.1	2.040
6	0.96	30.84%	0.1	2.010
7	0.96	30.53%	0.1	1.980
8	0.96	30.23%	0.1	1.950
9	0.96	29.92%	0.1	1.921
10	0.96	29.63%	0.1	1.892

Determination of emission reductions by ICS installed during year y

$$ER_{y,i,j} = B_{y,savings,i,j} \times NCV_{wood\ fuel} \times f_{NRB,y} \times (EF_{wf,CO_2} + EF_{wf,non\ CO_2}) \times N_{y,i,j} \times 0.95$$

Where

$$NCV_{wood\ fuel} = 0.0156 \text{ TJ/tonne}$$

$$f_{NRB,y} = 0.90$$

$$EF_{wf,CO_2} + EF_{wf,non\ CO_2} = 112 + 26.23 = 138.23 \text{ tCO}_2/\text{TJ}$$

Example of calculation:

If $y=2$,

$$ER_{y,i,j} = 2.132 \times 0.0156 \times 0.90 \times 138.23 \times 225,000 \times 0.95 = 884,278 \text{ tCO}_2$$

Year	$B_{y,savings,i,j}$	$NCV_{wood\ fuel}$	$f_{NRB,y}$	$EF_{wf,CO_2} + EF_{wf,non\ CO_2}$	$N_{y,i,j}$	$ER_{y,i,j}$
1	2.163	0.0156	0.90	138.23	1,12,500	448,625

2	2.132	0.0156	0.90	138.23	2,25,000	884,278
3	2.101	0.0156	0.90	138.23	3,37,500	1,307,153
4	2.070	0.0156	0.90	138.23	4,50,000	1,717,441
5	2.040	0.0156	0.90	138.23	4,05,000	1,523,040
6	2.010	0.0156	0.90	138.23	3,64,500	1,350,548
7	1.980	0.0156	0.90	138.23	3,28,050	1,197,506
8	1.950	0.0156	0.90	138.23	2,95,245	1,061,729
9	1.921	0.0156	0.90	138.23	2,65,721	941,276
10	1.892	0.0156	0.90	138.23	2,39,148	834,425

Year	Estimated baseline emissions or removals (tCO ₂ e)	Estimated project emissions or removals (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated net GHG emission reductions or removals (tCO ₂ e)
Year 1 27-September -2023 to 26-September -2024	-	-	-	448,625
Year 2 27-September -2024 to 26-September -2025	-	-	-	884,278
Year 3 27-September -2025 to 26-September -2026	-	-	-	1,307,153
Year 4 27-September -2026 to 26-September -2027	-	-	-	1,717,441
Year 5 27-September -2027 to 26-September -2028	-	-	-	1,523,040
Year 6 27-September -2028 to 26-September -2029	-	-	-	1,350,548
Year 7	-	-	-	1,197,506

27-September -2029 to 26-September -2030				
Year 8 27-September -2030 to 26-September -2031	-	-	-	1,061,729
Year 9 27-September -2031 to 26-September -2032	-	-	-	941,276
Year 10 27-September -2032 to 26-September -2033	-	-	-	834,425
Total	-	-	-	11,266,021

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	$f_{NRB,y}$
Data unit	Fraction
Description	Fraction of woody biomass saved by the project activity during year y that can be established as non-renewable biomass
Source of data	Official statistics or reports or peer-reviewed literature (UN Data & FAO reports)
Value applied	0.90
Justification of choice of data or description of measurement methods and procedures applied	This parameter shall be determined ex-ante. C4 EcoSolutions (Pty) Ltd was appointed as third party to study and derive the f_{NRB} value for Ghana. The value has been calculated in accordance with Tool30 ver4.0; using latest available versions of Food and Agriculture Organization (FAO) report, UN Data and other publicly available data that have been published by reliable sources.
Purpose of Data	Calculation of emission reductions
Comments	The report of f_{NRB} will be made available to VVB during the validation.

Data / Parameter	$NCV_{wood\ fuel}$
Data unit	TJ/tonne
Description	Net calorific value of the non-renewable woody biomass that is substituted or reduced
Source of data	2006 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy, Chapter 1 Introduction
Value applied	0.0156
Justification of choice of data or description of measurement methods and procedures applied	IPCC default value
Purpose of Data	Calculation of emission reductions
Comments	No Comments

Data / Parameter	$EF_{wf,CO2}$
Data unit	tCO ₂ /TJ
Description	CO ₂ emission factor for the use of wood fuel in baseline scenario
Source of data	2006 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy, Chapter 2 Stationary Combustion
Value applied	112
Justification of choice of data or description of measurement methods and procedures applied	IPCC default value
Purpose of Data	Calculation of emission reductions
Comments	No Comments

Data / Parameter	$EF_{wf,non\ CO2}$
Data unit	tCO ₂ /TJ

Description	Non-CO ₂ emission factor for the use of wood fuel in baseline scenario
Source of data	2006 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy, Chapter 2 Stationary Combustion
Value applied	26.23
Justification of choice of data or description of measurement methods and procedures applied	IPCC default value
Purpose of Data	Calculation of emission reductions
Comments	No Comments

Data / Parameter	η_p
Data unit	Fraction
Description	Efficiency of project stove at the start of project activity.
Source of data	Manufacturer's specification/ / third party efficiency test report.
Value applied	0.345
Justification of choice of data or description of measurement methods and procedures applied	This parameter shall be determined ex-ante
Purpose of Data	Calculation of $\eta_{new,y,i,j}$
Comments	No Comments

5.2 Data and Parameters Monitored

Data / Parameter	$N_{y,i,j}$
Data unit	Number
Description	Number of project devices of type i and batch j operating during year y

Source of data	Monitoring Survey / Registration records
Description of measurement methods and procedures to be applied	Measured directly or based on a representative sample. Sampling standard shall be used for determining the sample size to achieve 90/10 confidence precision according to the latest version (v9.0) of Standard for sampling and surveys for CDM project activities and programme of activities.
Frequency of monitoring/recording	At least once every two years
Value applied	For ex-ante emission reduction calculation, it is assumed that the project will distribute up to 500,000 ICS.
Monitoring equipment	Monitoring survey / Registration records
QA/QC procedures to be applied	The unique reference number of each stove shall be logged in the monitoring database showing the total number of stoves. Data from the sampling plan will be collected in each monitoring period by trained project staff and applied in the emission reduction calculations. Internal cross-checks by the Project Proponent will be undertaken as Quality Check.
Purpose of data	Calculation of emission reductions
Calculation method	Proportion of operational stoves obtained from the survey is multiplied by the total commissioned stoves to arrive at this value
Comments	As per applied methodology, Project Proponent needs to conduct the monitoring survey at least once every 2 years (biennial). However, the PP may conduct monitoring survey annually (at least once every year)

Data / Parameter:	η_{old}
Data unit:	Fraction
Description:	Efficiency of baseline stove
Equations	4
Source of data:	Default value: 0.1
Description of measurement methods and procedures to be applied:	A default value of 0.1 shall be used if baseline device is a three-stone fire using firewood (not charcoal), or a conventional device with no improved combustion air supply or flue gas ventilation, that is without a grate or a chimney.
Frequency of monitoring/recording:	Fixed for each individual household at the time of project implementation.

QA/QC procedures to be applied:	Not applicable
Purpose of data:	Calculation of emission reduction
Calculation method:	-
Comments:	A default value of 0.1 shall be used and is going to remain fixed throughout the crediting period, as the project intends to target only those households that are using the three stone fire or a conventional device with no improved combustion air supply or flue gas ventilation, that is without a grate or a chimney.

Data / Parameter	$\eta_{new,y,i,j}$																
Data unit	Fraction																
Description	Efficiency of the improved cookstove type i and batch j determined through water boiling test (WBT) during year y . Alternatively, efficiency may be determined using equation 5 of VMR0006 v1.1.																
Source of data	Calculation																
Description of measurement methods and procedures to be applied	To adopt Option V given in the methodology: “Efficiency of the improved cookstoves to be estimated using equation 5 above where loss in efficiency per year is calculated, and therefore this parameter does not need to be monitored”																
Frequency of monitoring/recording	Annually																
Value applied	For ex-ante calculation, the value below is applied. <table border="1"> <thead> <tr> <th>Year (y)</th><th>$\eta_{new,y,i,j}$</th></tr> </thead> <tbody> <tr><td>1</td><td>32.43%</td></tr> <tr><td>2</td><td>32.11%</td></tr> <tr><td>3</td><td>31.78%</td></tr> <tr><td>4</td><td>31.47%</td></tr> <tr><td>5</td><td>31.15%</td></tr> <tr><td>6</td><td>30.84%</td></tr> <tr><td>7</td><td>30.53%</td></tr> </tbody> </table>	Year (y)	$\eta_{new,y,i,j}$	1	32.43%	2	32.11%	3	31.78%	4	31.47%	5	31.15%	6	30.84%	7	30.53%
Year (y)	$\eta_{new,y,i,j}$																
1	32.43%																
2	32.11%																
3	31.78%																
4	31.47%																
5	31.15%																
6	30.84%																
7	30.53%																

	<table border="1"> <tr> <td>8</td><td>30.23%</td></tr> <tr> <td>9</td><td>29.92%</td></tr> <tr> <td>10</td><td>29.63%</td></tr> </table>	8	30.23%	9	29.92%	10	29.63%
8	30.23%						
9	29.92%						
10	29.63%						
Monitoring equipment	Not Applicable as it is calculated.						
QA/QC procedures to be applied	Not Applicable as it is calculated.						
Purpose of data	Calculation of emission reductions						
Calculation method	Calculation to be performed using equation below: $\eta_{new,y,i,j} = \eta_p \times (DF_n)^{y-1} \times 0.94$ $= 0.345 \times (0.99)^{1-1} \times 0.94$ $= 0.3243$						
Comments	No comments						

Data / Parameter	$B_{y=1,new,i,j,survey}$
Data unit	Tonnes/device/year
Description	Quantity of woody biomass used by project devices in tonnes per device of type i
Source of data	Monitoring survey
Description of measurement methods and procedures to be applied	Minimum sample size of each type i and batch j should be in line with the latest version of Standard for sampling and surveys for CDM project activities and programme of activities or guidelines provided in methodology Section 8.4 option (b). Determined in the first year of the introduction of the devices (e.g. during the first year of the crediting period, $y=1$) through measurement campaigns at representative households and/or sample survey. Sample surveys to estimate this parameter, that are solely based on questionnaires or interviews (i.e. that do not implement measurement campaigns) may only be used if the following conditions are satisfied. (i) Baseline cookstoves have been completely decommissioned and only improved cookstoves are exclusively used in the project households; (ii) If multiple devices are used in the project, it is possible from the results of the survey questions to clearly differentiate the quantity of firewood being used by each device. In other words, if more than one device, or another device that consumes firewood, are in use in project households, then the sample

	survey needs to distinguish the quantity of firewood used by the project device and the other devices that use firewood.
Frequency of monitoring/recording	Determined in the first year of project implementation
Value applied	For ex-ante calculation, the value is assumed as 2.64 kg/device/day or equal to 0.96 tonnes/device/year. This is assumed based on the literature review (details provided in ER sheet), to be confirmed during first Monitoring Period.
Monitoring equipment	Monitoring survey using weighing scale
QA/QC procedures to be applied	Survey to be conducted as per requirements of the methodology. Monitoring equipment will be calibrated as per the manufacturer 's guidelines.
Purpose of data	Calculation of emission reductions
Calculation method	Mean woody biomass consumption of sample stoves to be applied across entire population.
Comments	"Actual value of $B_{y=1,new,i,j,survey}$ will be determined in the first year of the implementation of the project through a sample survey."

Data / Parameter	Life Span
Data unit	Number of years
Description	The operating lifetime of the project device. The life span should be reported if the methodology equation 5 is adopted to determine the project stove efficiency
Source of data	Manufacturer's specification
Description of measurement methods and procedures to be applied	TLC-CQC Rocket Stoves will be manufactured to match the fixed design specification. This will be achieved using brick molds of specified dimensions to make bricks to be used for stove construction locally. This will ensure that each stove that is built at individual end user household measures exactly same as the dimensions specified by the manufacturer. Post/during construction, in person training will be provided to end users on use, care, and upkeep of these stoves. PP will conduct periodic audits and surveillance of the stoves that will be distributed under the project activity to ensure their proper functioning throughout the project lifetime. This along with spot audits and after installation maintenance services, will ensure that the project

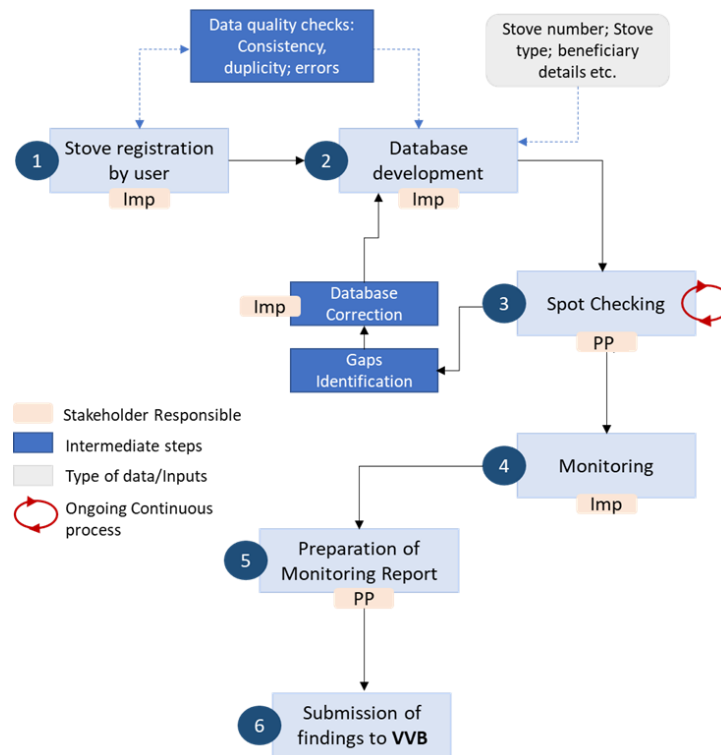
	stoves continue to work at efficiencies as specified by the manufacturer.
Frequency of monitoring/recording	Once at the time of project stove installation
Value applied	10
Monitoring equipment	This parameter will be referred from the Manufacturer's specification
QA/QC procedures to be applied	This parameter will be referred from the Manufacturer's specification
Purpose of data	Calculation of emission reductions
Calculation method	No calculation was used for this parameter
Comments	No Comments

5.3 Monitoring Plan

The Monitoring Plan applied in this project involves several key elements that ensure that the PP and/or project Implementer have high-quality, unbiased, and reliable information regarding the performance of the project in terms of implementation and outcomes, and for the purposes of calculating Verified Carbon Units (VCUs) following VCS methodology (VMR0006 version 1.1) on the basis of the amount of non-renewable biomass saved by the ICS in the project activity. The key elements are the following:

- Data collection procedures
- Distribution and Monitoring Database
- Spot Checking of ICS (ongoing)
- Sample Plan for the Monitoring Survey
- Data Quality, Consistency and Duplication Checks
- Monitoring Reporting

The below flow-chart illustrates the roles and responsibilities of the parties during the implementation of the monitoring plan for the project. In the below flowchart, the project implementer is abbreviated to "Imp" and can be the PP or another party authorized by the PP.



Roles and Responsibilities of stakeholders during the implementation of the monitoring plan

Below is the description of the above steps on the flow-chart.

- 1. Implementor: User registers stove:** Project implementer will collect/receive the necessary information requested in the Registration process from the user. Means of collecting this information may be through a physical registration card filled by project Implementor staff, retailers, end-users, or partner organization's staff, or through the use of ICTs or SMS. Project Implementers' staff shall double check the accuracy of information provided, and request for field staff additional clarifications if needed. Following is the information that will be captured from each end-user at the time of registration of ICS during the project implementation, to be eligible for inclusion in the project activity:
 - Date of distribution/installation of ICS.
 - Geographic area/address of recipient household.
 - GPS coordinates of the installed project device.
 - ICS Model/type of project technology distributed.
 - Quantity/number of project device that will be distributed.
 - End-User name and telephone number (if available).
 - Unique identification alphanumeric ID for each ICS that will be distributed.
- 2. Implementor: Data logged into database:** Project implementer's trained staff will input the data in the database either manually (if data collected from physical registration card) or this

will be automatically input if data was collected using ICTs or SMS. Project implementer staff shall double check the information included on the database and check for duplications. Any duplicate information shall be investigated, and errors corrected or excluded from the database if it is a true duplicate entry.

3. **Project Proponent: Spot- checking (ongoing):** Project Proponent has a procedure for internal auditing called spot-checking, where field staff randomly selects households to ensure proper functioning of the project stoves throughout the project lifetime and to ensure that the project stoves continue to work as efficient as specified by the manufacturer. Field staff of project proponent randomly select units included in the database and visit or contact the stove owners to cross-check the information on the database with the factual evidence in the field. Any inconsistencies found (e.g., change in the address of a user) gets updated on the database, and in case, ICS are found to be no longer in use, they get clearly marked as such and is excluded from emission reductions calculations.

During the spot check, if any of the project stoves was found to be in poor working condition like cracked/damaged or with missing metal parts, the stove owner repairs the stove as they have been trained during installation. If the stove owners are unsuccessful in repairing, the field staff informs the implementer to provide the necessary assistance and the missing metal parts are reported to the management of implementer who then either provides the parts from their existing stocks or request the same to project proponent who procured those from their suppliers. Implementing partner then provides those parts to the stove owners. This is a continuous process.

4. **Implementor: Monitoring:** Project implementer will follow the requirements as per PD to collect the necessary information for a monitoring report. The project proponent will assess all monitoring data and produce a monitoring report for the VVB to verify corresponding to the preceding monitoring period. This report will present the data relating to the emission reductions generated by those PAIs during the monitoring period.
5. **Project Proponent: Preparation of monitoring report:** the project implementers or the PP will prepare the final monitoring report to be provided to the VVB for verification of emission reductions. A copy of the monitoring report will remain with the Project Proponent.

The PP will coordinate and manage each project Implementer and assist them in implementing each element of the monitoring plan. The monitoring plan shall be elaborated in accordance with the Sampling Plan below. Also, regular audits conducted throughout the project lifetime ensures proper functioning of stoves and determination of the lifespan of the same.

Sampling Plan

As per the Guideline for Sampling and Surveys for CDM Project Activities and Programme of Activities, version 04.0, the sampling plan is the following:

A. Sampling design

Due to the large number of ICS envisioned to be distributed under the project, it is not economically feasible to monitor each individual ICS unit distributed. Therefore, representative sampling shall be undertaken for each type of ICS. The sampling plan to be designed in line with the requirements of the Standard for Sampling and Surveys for CDM Project Activities and Programme of Activities, version 09.0.

Samples will be drawn from ICS recorded in the project database administered by the project proponent. A detailed explanation of this database is found in Section 1.11 of project description.

i. Objectives and Reliability Requirements

The objective is to obtain an unbiased and reliable estimate of the proportion or mean value of the following key variables over the course of the crediting period and with 90/10 confidence/precision (as per applied methodology VMR0006; version 1.1).

Monitored Parameters:

The project involves introduction of improved cookstoves as energy efficiency measure

No.	Monitoring parameters	Sampling parameters	Parameter type	Monitoring frequency
1	$N_{y,i,j}$ Number of project devices of type i and batch j operating during year y	Proportion of ICS still in operation	Proportion	At least once in two years.
2	$B_{y=1,new,i,j,survey}$ Quantity of woody biomass used by project devices in tonnes per device of type i and batch j	Daily consumption of woody biomass per ICS	Mean value	Determined in the first year of project implementation

ii. Target Population

The target population will be the complete set of appliances (ICS) deployed under the project.

iii. Sampling Frame

It is expected that the geographical locations do not have influence on the parameters of interest that is $N_{y,i,j}$ & $B_{y=1,new,i,survey}$. Therefore, these parameters can be assumed to be highly homogeneous regardless of how the end user group and distribution/installation location is defined since the end users shall always be households and geographical location is restricted to the project boundary. The Sampling Frame for individual parameters have been defined in the following paragraphs:

The sample frame refers to all the information sources on the Database. The SMS/ICT system is the primary mechanism for data collection for newly distributed ICS and the Monitoring Survey (which includes a household questionnaire and visual inspection of ICSs) that will be used throughout the lifetime of the project. The SMS/ICT data is used to populate the stoves Database and the Monitoring Survey follows the “Standard for Sampling and Surveys for CDM Project Activities and Programme of Activities”; version 09.0.

All the TLC-CQC rocket stoves distributed under this project shall be grouped together to create a Primary Sampling Unit which is homogenous. The procedure to determine the sample of households will ensure that they adequately represent the broader project population, minimizing

sampling error. Using, a 90 per cent confidence level, and a 10 per cent margin of error, a random sample will be selected from the target population.

a. Sampling frame for proportion of ICS still in operation ($N_{y,i}$)

The first step is to identify the Primary Sampling Units.

Factors such as model of the ICS do not effect parameter $N_{y,j}$. Therefore Primary sampling unit in case of parameters $N_{y,j}$ is independent of model of stove.

b. Quantity of woody biomass used by project devices in tonnes per device ($B_{y=1,new,i,survey}$)

The Parameter $B_{y=1,new,i,survey}$ shall vary in accordance with its model. Hence for this parameter, the Primary Sampling Unit shall be defined as the group of ICSs of the same model. If two different ICS models being implemented in the same project – this will form two Primary Sampling Units. The value will be determined in the first year of the introduction of the devices through the monitoring survey.

Under this project upto two TLC-CQC Rocket Stoves will be installed in each household, which are classified as Project stove 1 and project stove 2. At the time of monitoring survey, field staff record the same in survey app. Therefore, firewood consumed for each project stove and baseline stove can be distinguished clearly.

Calibration of weighing scales use for measuring the fuel wood will be done in household before start using on site.

iv. Sampling Method

The sampling method for the monitored parameters $N_{y,i,j}$, and $B_{y=1,new,i,survey}$ is Simple Random Sampling and samples will be randomly selected from the primary sampling units as illustrated above. To ensure a random selection of ICS, random number generators shall be applied. Each ICS in the target population is uniquely identifiable by its unique ID number. Each ICS can thus be allocated a Sample Selection Number in each monitoring period, starting at 1 and increasing up to the total number of ICS in the Database for that pre-defined sampling frame. Applying the random number generators, the ICS can then be randomly chosen from the defined population up to the required sample size as calculated by the project proponent.

To determine the parameters, sampling will involve the following approaches (outcome in brackets):

$N_{y,i,j}$	Visual inspection of the premises to see if ICS is operational and in use. Interview with end user if required to verify that ICS is still in use (Yes/No)
$B_{y=1,new,i,j,survey}$	Quantity of woody biomass used by project devices in tones per device per year (First year of installation).

v. Sample Size

For the estimation of the proportion or mean value of the parameters investigated, the minimum sample size for each sample frame has to achieve the 90/10 confidence/precision.

The procedure to determine the sample of households will ensure that they adequately represent the broader project population, minimizing sampling error. Using a 90 per cent confidence level, and a 10 per cent margin of error, random samples will be selected from each Primary Sampling Unit. The parameter number of stoves still in operation during the monitoring period as determined by the monitoring survey ($N_{y,j}$) will be estimated through sampling annually/biennially and the parameter Quantity of woody biomass used by project devices in tonnes per device $B_{y=1,new,i, survey}$ shall be estimated through sampling within the first year of project installation. $B_{y=1,new,i, survey}$ can be coupled with $N_{y,j,j}$ for the first year. Of the two parameters to be monitored, $N_{y,i,j}$ is proportion/percentage parameter and $B_{y=1,new,i, survey}$ is a mean value parameter.

In order to calculate the required sample size estimates, values for the proportions, mean values, and standard deviations are required. As per Guideline for Sampling and surveys for CDM project activities and programmes of activities, version 04.0, there are different ways available to obtain the estimates of the parameter of interest:

- (a) Refer to the result of previous studies and use these results;
- (b) In a situation where information from previous studies is not available, a preliminary sample as a pilot could be conducted and use that sample is used to provide the estimates;
- (c) Use best guesses based on the researcher's own experiences.

For the registration purpose of project, option (c) as stated above shall be applied. For the following monitoring periods, the estimates can be adjusted taken the results of the previous monitoring period(s) into account or the result from recent pilot study which is conducted after the previous monitoring periods.

Parameter $N_{y,i,j}$:

To estimate the sample size for the parameter $N_{y,i,j}$ the following equation is used:

$$n \geq \frac{1.645^2 N \times p \times (1 - p)}{(N - 1) \times 0.1^2 \times p^2 + 1.645^2 p \times (1 - p)}$$

Where:

- | | |
|-------|--|
| n | = Sample size |
| N | = Population size (Total number of households/ICS) |
| p | = Expected proportion |
| 1.645 | = Represents the 90% confidence required (In the case of 95% confidence, 1.96 shall be used) |
| 0.1 | = Represents the 10% relative precision |

The following assumptions are made to exemplify the sample size calculation for the parameters.

1. An overview of the estimated sample sizes for a hypothetical population of 100,000 ICS units applying a level of 90/10 is provided below. It is likely that all the sample frames for each parameter will include fewer than 100,000 ICS in the first monitoring period, so this is a conservative approach. Hence, population size, N, is taken as 100,000 households/ICS (Assuming one ICS for one household).
2. It is expected at least 80% of ICS still in operation, hence the expected proportion p is taken as 0.8.

Sample size calculation:

The calculation of the required sample size for each parameter in the first monitoring period is illustrated below. In all cases a conservative approach is taken, however if for any parameter the required confidence/precision is not met then the PP will randomly select an additional sample and collect further data from this sample to ensure the pooled data meet or exceed the required thresholds.

Based on the above assumptions, the resulting sampling size is calculated as:

$$n \geq \frac{1.645^2 \times 100,000 \times 0.8(1 - 0.8)}{(100,000 - 1) \times 0.1^2 \times 0.8^2 + 1.645^2 \times 0.8(1 - 0.8)} = 67.61$$

Therefore, in this case a sample size of 68 is to be sampled from each primary sampling unit.

Parameter $B_{y=1, \text{new}, i, \text{survey}}$:

For the purposes of determining sample size in the first monitoring period, the performance of ICS is characterized by the range of likely mean wood consumption and the likely values of SD relative to the mean, according to the type of ICS. The ICS models that are manufactured in modern factories tend to be very highly efficient (30-50% thermal efficiency) and have been designed to meet stringent efficiency specifications, so the standard deviation is expected to be relatively low.

To estimate the sample size for parameter $B_{y=1, \text{new}, i, \text{survey}}$ the following equation is used:

$$n \geq \frac{1.645^2 NV}{(N - 1) \times 0.1^2 + 1.645^2 V}$$

Where:

$$V = \left(\frac{SD}{mean} \right)^2$$

n = Sample size

N = Population size (assumed 0.38 as example)

- mean = Expected mean of ICS thermal efficiency (assumed 0.076 as example)
- SD = Expected standard deviation
- 1.645 = Represents the 90% confidence required
- 0.1 = Represents the 10% relative precision

Based on the above assumptions, the sample size calculation would be

$$n \geq \frac{1.645^2 \times 100,000 \times \left(\frac{0.076}{0.38}\right)^2}{(100,000 - 1) \times 0.1^2 + 1.645^2 \times \left(\frac{0.076}{0.38}\right)^2} = 10.82$$

The resulting sample size based on the above equation is smaller than 30, since $B_{y=1, \text{new}, i, \text{survey}}$ is a numeric mean value (i.e. not a proportion or percentage) the Student's t-distribution shall be used as per paragraph 14 of "Standard for Sampling and Surveys for CDM Project Activities and Programme of Activities, version 09.0.

The sample size for parameter $B_{y=1, \text{new}, i, \text{survey}}$ under t-distribution is referred to the equation below:

$$n = \left(\frac{t_{n-1} \times SD}{0.1 \times \text{mean}} \right)^2$$

Where t_{n-1} is the value of the t-distribution for 90% confidence when the sample size is n. Since the sample size is not known yet, the first step is to use the value for 90% confidence when the sample is large, i.e., 1.645 and then redefine the calculation.

$$n = \left(\frac{1.645 \times 0.076}{0.1 \times 0.38} \right)^2 = 10.82$$

Thus, n is rounded up to 11.

The calculation now needs to repeat using t-value for 90% confidence and n = 11

$$n = \left(\frac{2.228 \times 0.076}{0.1 \times 0.38} \right)^2 = 19.86$$

And n is rounded to 20.

The calculation now needs to repeat using t_{n-1} value for n = 20. The process should be iterated until there is no change to the value of n.

T_{20-1}	2.093
n=	17.52
Round up	18

t_{18-1}	2.110
$n=$	17.81
Round up	18

The repeated calculation shows that $n = 18$. Thus, the sample size to be sampled from each sampling unit is 18.

Since parameters $N_{y,j,j}$, and $B_{y=1,new,i, survey}$ share the same sampling units, PP may choose to have one common survey for these two parameters with largest number of sample size between these two parameters being chosen. Sampling more than one parameter within the same sample (household) helps reduce travel needs for monitoring and the associated costs. At the same time this approach ensures the random selection of samples for every parameter.

It is a good practice to employ oversampling at the design stage, not only to compensate for any attrition, outliers or non-response associated with the sample, but also to prevent a situation at the analysis stage where the required reliability is not achieved, and additional sampling efforts would be required. The sample size shown above will be adjusted upwards to account for non-responses, PP shall determine the appropriate non-responses rate based on previous experience.

B. Data:

i. Field Measurements:

To monitor the number of stoves that continue to be in use ($N_{y,i,j}$), the data collected will be a representative number of stoves in the database that are in use for the monitoring period. The method of collecting data will be field surveys of required sample size of ICS users in the database. Data will be collected from the field surveys, entered in the database and included in the monitoring report.

For monitoring $B_{y=1,new,i, survey}$, the data collected will be representative number of stoves in the database that were distributed within one year of start date of project implementation. The method of collecting data will be field survey and frequency will be once during the entire crediting period.

The table below summarizes field measurement data requirements

Parameter	Timing (indicative)	Frequency as per methodology	Methods to be applied	Comments on seasonal fluctuation
$N_{y,i,j}$	Monitoring will likely occur at least once in every 12 months	At least once in two year.	Visits to the end user premises, visual inspection and interview with ICS end-user	Unlikely to be due to any seasonal fluctuation.
$B_{y=1,new,i, survey}$	Within 1 st year from the starting date of	Once. The value will be fixed for	Visits to the premises, visual inspection, measurement and	Unlikely. Seasonal fluctuation is not expected to affect this parameter.

	project implementation	entire crediting period	interview with ICS end-user.	However, in case the survey results show a difference, the same shall be taken into account in ER estimation.
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Soft copy of original stove purchase contracts, information collected at the time of registration or other means of acceptance by the users will be stored in the project office. A back-up of the project database will also be stored on an electric medium. All data monitored and required for verification and issuance will be kept for two years after the end of the crediting period or the last issuance of credits for the project activity, whichever is later.

ii. Quality Assurance/Quality Control

The PP will apply measures to ensure the required confidence/precision for each sampled parameter is met, allowing for non-response and the possible removal of outliers from the sample, as part of a Quality Control/Quality Assurance system. The choice of measure applied to each parameter will depend on the cost of each data collection approach and logistics required. The PP will determine the most effective measure for each parameter from the following list (illustrated using a required sample size of 20 and an effect of non-response of 2 to 4 ICS):

- Oversampling: Randomly draw a sample of minimum 24 ICS and collect data from each
- Buffer Group: Randomly draw a sample of at minimum 24 ICS and collect data from only 22 ICS. If this would not result in the required sample size data would be collected from the additional 2 ICS that were selected in the sample.
- Draw an additional sample: Randomly draw a sample of 22 ICS and collect data from these. If the required sample size is not achieved, an additional sample of 2 elements will be drawn and included in the sample.
- Use lower confidence bound (of $N_{y,i,j}$, $B_{y=1,new,i,survey}$) or, with a conservative approach according to the parameter definitions,

The PP may choose to stop monitoring a particular parameter once the required level of confidence/precision has been reached, as long as the calculated minimum number of samples has been achieved. As an example, the following steps could logically be followed for the case of applying a 30% buffer:

- Visit first 10% of premises required for the 30% buffer. If the number of responses is sufficient to achieve the required reliability level, then stop sampling.
- If step 1 is not sufficient to achieve the required reliability level, then visit the next 10% of premises (increases the additional sampling to 20% of the 30% buffer). If this additional sampling is sufficient, then stop sampling.
- If step 2 is not sufficient to achieve the required reliability level, then complete the final 10% of the additional sampling buffer (bringing the total to 30%).

The sampling plan has the following procedures in place to ensure good quality data. The PP will ensure that field personnel have reviewed, understand and have signed the monitoring plan, including provisions for maximizing response rates, documenting out-of-population cases, refusals

and other sources of non-response. A quality control and assurance strategy will be documented. Quality control and assurance strategies include addressing non-sampling errors, such as non-response or bias from interviewer. The monitoring plan will explain how to properly survey households to prevent bias from interviewer. In the case a household refuses to participate, another household will be chosen at random. To reduce interviewer bias, good questionnaire design and well-tested questionnaires will be used.

The calculation of the sample size will be carried out using estimates for parameter proportions, mean values and standard deviations, as the actual characteristics of the population/sampling frame are unknown. In order to ensure the quality of the sampling results, the PP can draw on the provisions for reliability calculations as provided by the Guidelines for Sampling and Surveys in CDM Project Activities and Programme of Activities (version 04). In the event that the sampling results do not fulfil the required level of confidence and precision, the PP can undertake additional samples. If the reliability is still not sufficient after additional samples or other measures, the sampling may be repeated with an increased sample size. Alternatively, the PP may choose to apply the lower bound or higher bound according to the more conservative approach.

iii. Data archiving

Hard copies of the surveys will be kept and the registration database will have back up. Original stove purchase contracts or other means of acceptance by the users will be stored in the main office for the coordinating entity. A back-up of the registration database will also be stored on an electronic medium by the PP. All data monitored and required for verification and issuance will be kept for two years after the end of the crediting period or the last issuance of VCUs for the project activity, whichever is later.

iv. Analysis

The PP will manage a project database that includes the following data that can be directly attributable to each ICS within the project activity, thereby allowing unambiguous determination of the emission reductions attributable the project:

- A list of households participating in project activity, including name, community/location, distribution/ installation date and unique serial number;
- Where replacements are made, assurance that the efficiency of the new ICS is similar to the specified.

Data obtained from the samples will be used to estimate proportions and mean values for the parameters described above. The values will then be factored into the emissions reduction calculations and result in the request for issuance of VCUs. The stoves that are not in use will be excluded from emissions reductions calculations and will not be counted towards the total number of ICS in operation during the monitoring period.

C. Implementation

Sampling for the purpose of emission reduction calculation and elaboration of the monitoring report will occur at the end of each monitoring period. This sampling will be conducted by trained personnel either part of the project Implementer or PP team, or an experienced third-party entity. The maximum length of one monitoring period will be two years (duration, not calendar years). The project Implementer will be responsible for managing household data collection and entry into the project database. Field personnel will receive training on how to properly deal with surveying

techniques and reduce errors and sign a document certifying that there is no conflict of interest of those involved in data collection and analysis. If there is conflict of interest, the personnel will not be allowed to participate in data collection and analysis. The project database will record the start and end dates of each monitoring period and record the emission reductions attributable to each monitoring period. Appropriate record keeping procedures will be implemented to ensure that each monitoring period data set can be transparently attributed to its corresponding project, preventing any occurrences of double counting. An internal review of the project database will be able to determine the status of project—the duration of previous monitoring periods, the households delivering monitoring data, and current verification activities.

Assessment for Leakage

According to methodology VMR0006, version 1.1, section 8.3; Leakage shall be considered as default 0.95 in accordance with Section 5.4 of AMS-II.G. therefore default value of leakage as 0.95 will be applied.

Disposal of Low Efficiency Appliances and Use of Baseline Stoves

When an ICS is installed, the end user receives information explaining that conventional open fire appliance must no longer be used. Follow-up meetings with end users will ensure that those who have received an ICS are using it properly and that the conventional open fire is no longer in use.

Monitoring Reporting

The PP will assess all monitoring data and produce one or two monitoring reports for the project for the VVB to verify corresponding to the preceding monitoring period the project. This report will present the data relating to the emission reductions generated by the project at the time of the monitoring period.

APPENDIX

Appendix A- Local Stakeholder Consultation

LSC Public notice

Daily Graphic, Friday, March 17, 2023

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News

GJA, 2 others sign MoU to train female journalists

By Zolok Kwame Gyasi

THE Ghana India ICT Alliance Centre of Excellence in ICT (GIA-ACE), an agency under the Ministry of Communications and Digitalisation, has signed a memorandum of understanding (MoU) with the Ghana Initiative for Women in Media Africa (GIWMA) to train female journalists on digital skills.

The MoU was signed at this year's Females in Tech (FEMTECH) Conference organised by GIWMA on March 8, 2023, which coincided with the International Women's Day celebration.

The MoU aims to address the gender gap in digital skills among journalists, and empower more women to excel in the profession.

The FEMTECH Conference aims to inspire and empower women in the industry and create a supportive ecosystem to help them achieve their goals.

Benefits of MoU

At the signing of the MoU, the Executive General of GIWMA, Dr Collins Yehou-Adjei, expressed delight with the partnership to empower female journalists in the country with digital skills.

He said the media industry was rapidly changing, hence the need to equip journalists with the latest tools and techniques to better stay relevant and competitive.

"Through this partnership, we hope to bridge the gender gap in digital skills and create opportunities for women to advance in their careers," he noted, adding that "this partnership will benefit female journalists and the communities they serve."

Dr Yehou-Adjei said by equipping female journalists with the necessary ICT skills, "we can help to ensure that their voices are heard, and that important stories are told."

Awards

The Ministry of Communications and Digitalisation, United Nations Development Programme (UNDP), expressed the view that technology was made for women, adding that when given the tools to work with them, women could work successfully, efficiently, creatively and safely.

The event was also attended by some notable Ghanaian women, including Kwanmaria Doo, the Managing Director of Ghana Ghana Group, Dr Lucy Agyemang, Associate Dean of Engineering, Academic City University College, Science Dept Head, Dr Netteh Business Services for Tech Solutions Africa, Prof. Elina Telleh Kachwera, Dean of Engineering, University of Ghana, Mrs Eva Anakuwa-Poku, Deputy CEO, GTEC, tertiary students, students from several cycle institutions, ICT groups, and individuals from corporate Ghana.

Mr Obedson Ekechi, Prof. Kachumona, Dr Agyemang and the Chief Operations Officer and Co-Founder of Super Field AI Lab, Winfred Borne, and Nana Yao Kwaku, a prominent media practitioner at Joy FM, were honoured for their contributions to the industry.

* Linda Asante Aducci (right), Vice President of GIA, signing the MoU with Dr Collins Yehou-Adjei, the Director General of GIWMA.

CAREER OPPORTUNITY

GRAPHIC COMMUNICATIONS GROUP LIMITED

ASSISTANT ADVERTS SALES & MARKETING OFFICER (DIGITAL)

The group wishes to engage the services of an enthusiastic, energetic and results-oriented individual who will be responsible for actively marketing the group's digital brands and the sale of digital channels.

Key Accountabilities

- Manage existing client relationships, regularly visiting clients and agencies to generate sales and canvass for advertising opportunities.
- Actively seek new clients and sources of advertising revenues, developing and managing new relationships.
- Ensure that weekly advertising targets are met.
- Proactively provide feedback from clients to management on issues of client concern.
- Work with clients on developing advertising campaigns in terms of advising on correct publications and media vehicles.
- Work with clients on content and format of advertising campaigns.
- Promote Graphic's events, seeking sponsorship and promoting clients' active participation in events, including client placement, such as sign, banners, etc.
- Digital marketing and sale of digital channels.

Skills/Competencies Required

- Ability to implement efficient and accurate financial systems.
- Must be a team player with a high sense of commitment, honesty and integrity.
- Must possess good communication and negotiation skills and have a keen eye for detail.

Qualifications & Experience

- Must possess a good first degree in Marketing, CSMS qualification, Business Administration, Social Sciences or equivalent from a reputable tertiary institution and at least two (2) years of post-qualification working experience in Sales & Marketing.
- A minimum of HND in Marketing and at least four (4) years of post-qualification relevant working experience in Sales & Marketing.

Interested applicants who meet the above criteria may send applications, CVs and supporting documents to the address below.

Ag Head of HR
Graphic Communications Group Ltd.
P.O. Box 67142
Accra

Alternatively, you may email your application to:
recruitment@graphic.com.gh

Closing date for submission of applications: 31st March, 2023

Please note that only shortlisted applicants will be contacted.

CQuestCapital

INVITATION TO LOCAL STAKEHOLDER CONSULTATION ON CLEAN COOKSTOVES

C-Quest Capital is developing an improved cookstove program in Ghana. The proposed project will be developed under the Verified Carbon Standard (VCS) and the Sustainable Development Verified Impact Standard (SDI) to obtain SDI Verified Impact Standard (VCS) to increase the livelihood of the project. Through the distribution and installation of our energy-efficient cookstoves, we aim to reduce the use of non-renewable biomass for cooking and water heating for households, particularly in rural Ghana.

To better design the project, C-Quest Capital has scheduled a local stakeholder consultation to gather comments and suggestions.

Date: Monday, 3rd April 2023
Time: 12:00
Venue: Main Auditorium, ISSER Conference Facility, University of Ghana.

For more information about the event or to provide feedback, please contact C-Quest Capital at web@cg.cquestcapital.com or +233 66 696566.

All are cordially invited.

CQuestCapital

INVITATION TO LOCAL STAKEHOLDER CONSULTATION ON CLEAN COOKSTOVES

C-Quest Capital is developing an improved cookstove program in Ghana. The proposed project will be developed under the Verified Carbon Standard (VCS) and the Sustainable Development Verified Impact Standard (SD Vista) to obtain SD Vista labelled Verified Carbon Units (VCUs) to increase the feasibility of the project. Through the distribution and installation of our energy efficient cookstoves, we aim to reduce the use of non-renewable biomass for cooking and water boiling for households, particularly in rural Ghana.

To better design the project, C-Quest Capital has scheduled a local stakeholder consultation to gather comments and suggestions.

Date: Monday, 3rd April 2023

Time: 10am

Venue: Main Auditorium, ISSER Conference Facility, University of Ghana.

For more information about the event or to provide feedback, please contact C-Quest Capital at aadjabeng.ctr@cquestcapital.com
OR +233 53 0391988

All are cordially invited.

Appendix B- List of Abbreviations:

CDM: Clean Development Mechanism

CQC: C-Quest Capital LLC

GHG: Green House Gas

ICS: Improved Cookstoves

PD: Project Description

PP: Project Proponent

SDG: Sustainable Development Goals

SD ViSta: Sustainable Development Verified Impact Standard

TLC: Total Land Care

TSF: Three Stone Fire

VCS: Voluntary Carbon Standard

VCU: Voluntary Carbon Unit

VVB: Validation and Verification Body

UNFCCC: The United Nations Framework Convention on Climate Change

Appendix C - Scope 3 emissions notification(s)

Public Notice: <https://cquestcapital.com/latest/public-notices/>

Refer Public Notice for VCS Project 4161

Public Notice for VCS Project 4161

Installation of High Efficiency Wood Burning Cookstoves in Ghana

This is to bring to your kind notice that C-Quest Capital CR Stoves Private Limited is implementing a grouped project titled, "Installation of high efficiency wood burning cookstoves in Ghana" (VCS ID 4161). It involves distribution of improved cook stoves (ICS) to households in Republic of Ghana. For this purpose, we are procuring metal parts of TLC-CQC Rocket Stove from Ener-G-Africa Holding Ltd. We have also involved 3D Ecolutions and a CQC Direct Team as implementation partner for distribution of project stoves in Republic of Ghana.

Verified Carbon Units (VCUs) may be issued for the greenhouse gas emission reductions and removals for which C-Quest Capital CR Stoves Private Limited will be claiming carbon credits under VERRA. The ownership of these credits lies exclusively with C-Quest Capital CR Stoves Private Limited.

This notice intends to apprise you about the project, to avoid any potential risk of double claiming of Scope 3 emissions under this grouped project.